OPENSOURCESDRLAB DOCUMENTATION

B210-Kintex-7 Schematic

Product	B210-Kintex-7
Document	OSSDRLAB-B210-Kintex-7-SCH
Version	2.1
Date	2026-11-12

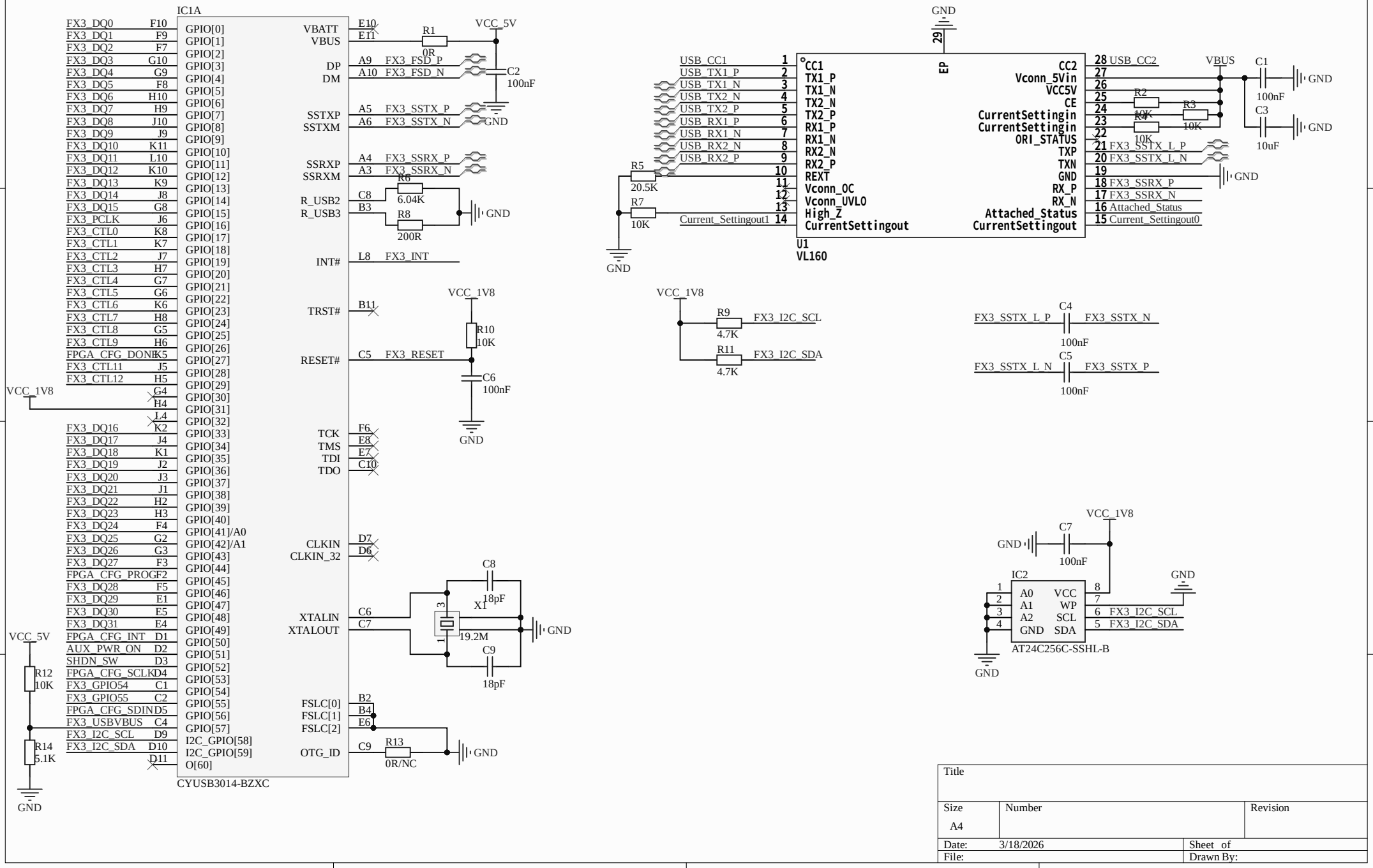
Circuit schematic (board revision V2.1) of the OpenSourceSDRLab B210-Kintex-7, a USRP B210-compatible SDR with an Analog Devices AD9361 and a Xilinx Kintex-7 XC7K325T FPGA. Exported from Altium Designer (A4); sheet 13 is blank.

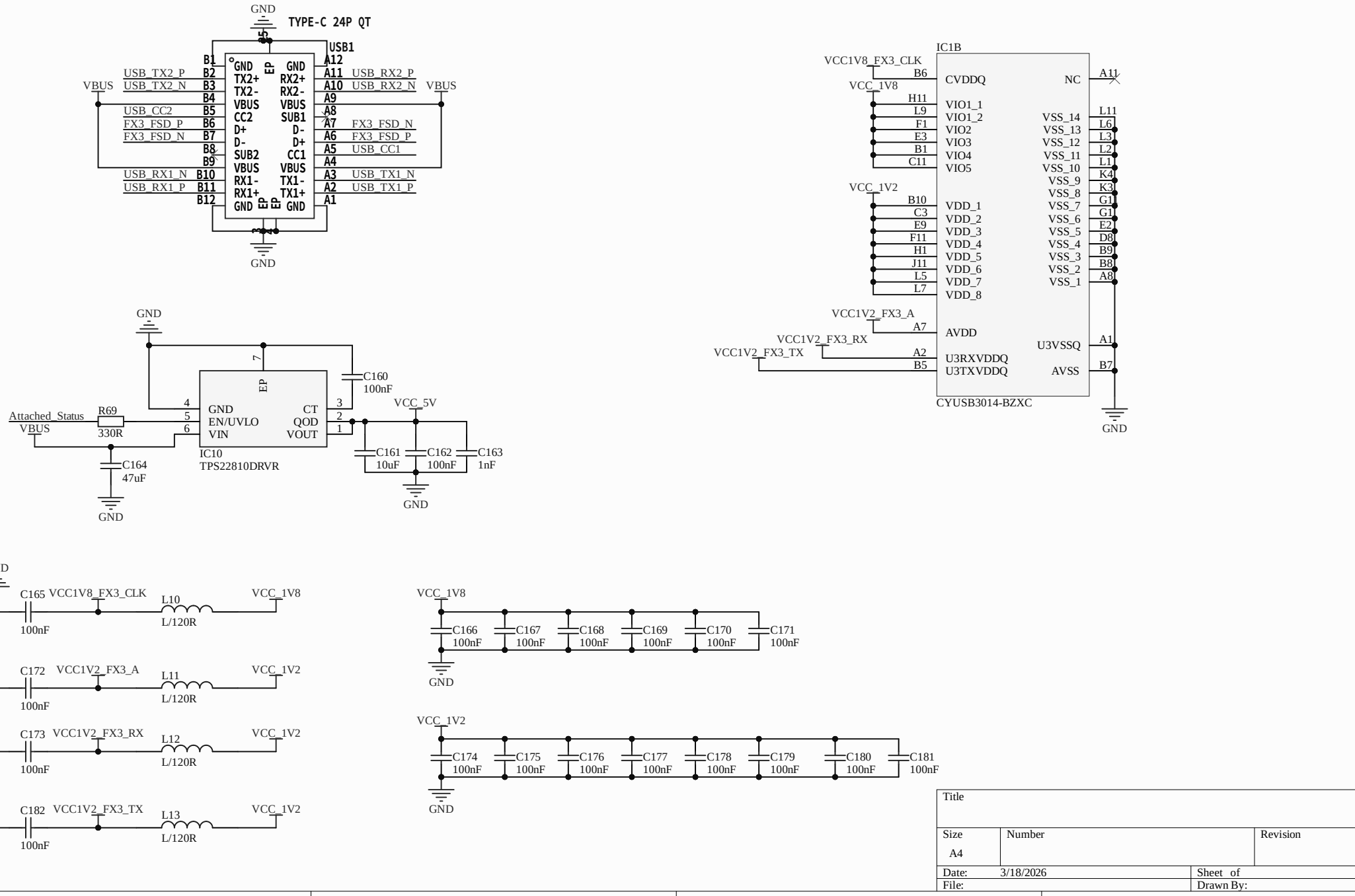
Sheet	Content
1–2	USB 3.0: CYUSB3014 (FX3) controller, GPIF II bus, Type-C connector and FX3 supply
3	Power: MP2143 step-down converters, TPS7A9201 low-noise LDOs
4–5	FPGA configuration, core supply (TPS53355), power pins and decoupling
6–7	FPGA HR/HP I/O banks: AD9361 interface, FX3 GPIF II, PLL, GPS, GPIO
8	AD9361 RF transceiver
9–10	RF front ends, channel 1 (TRX1, RX1) and channel 2 (TRX2, RX2)
11	Status LEDs, GPIO/JTAG FPC connector
12	Clocking: ADF4001 PLL, 10 MHz and PPS inputs, onboard GPS module

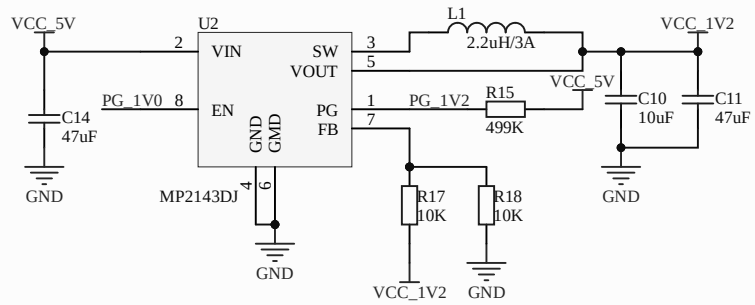
Product page: https://opensourcesdrlab.com/products/b210-kintex-7

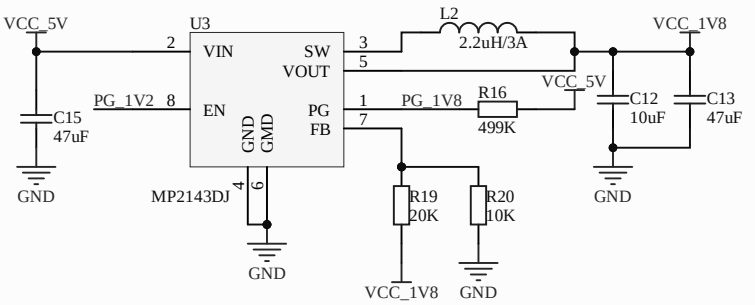
Questions about the hardware: support@opensourcesdrlab.com

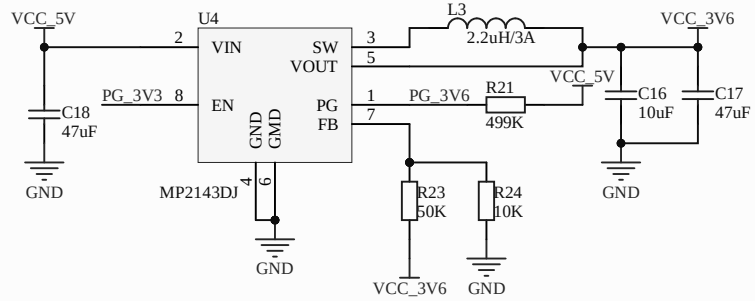
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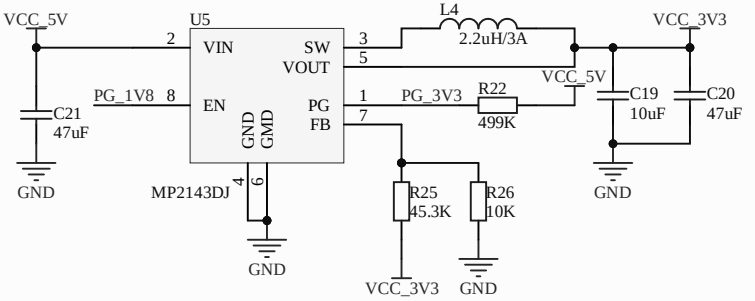

IC1A
CYUSB3014-BZXC
U1
VL160
IC2
AT24C256C-SSHL-B
VCC_5V
VCC_1V8
GND
R1
0R
C1
100nF
C2
100nF
C3
10uF
R2
10K
R3
10K
R4
10K
R5
20.5K
R6
6.04K
R7
10K
R8
200R
R9
4.7K
R10
10K
R11
4.7K
R12
10K
R13
0R/NC
R14
5.1K
C4
100nF
C5
100nF
C6
100nF
C7
100nF
C8
18pF
C9
18pF
X1
19.2M
FX3_DQ0
FX3_DQ1
FX3_DQ2
FX3_DQ3
FX3_DQ4
FX3_DQ5
FX3_DQ6
FX3_DQ7
FX3_DQ8
FX3_DQ9
FX3_DQ10
FX3_DQ11
FX3_DQ12
FX3_DQ13
FX3_DQ14
FX3_DQ15
FX3_PCLK
FX3_CTL0
FX3_CTL1
FX3_CTL2
FX3_CTL3
FX3_CTL4
FX3_CTL5
FX3_CTL6
FX3_CTL7
FX3_CTL8
FX3_CTL9
FPGA_CFG_DONE
FX3_CTL11
FX3_CTL12
FX3_DQ16
FX3_DQ17
FX3_DQ18
FX3_DQ19
FX3_DQ20
FX3_DQ21
FX3_DQ22
FX3_DQ23
FX3_DQ24
FX3_DQ25
FX3_DQ26
FX3_DQ27
FPGA_CFG_PROG
FX3_DQ28
FX3_DQ29
FX3_DQ30
FX3_DQ31
FPGA_CFG_INT
AUX_PWR_ON
SHDN_SW
FPGA_CFG_SCLK
FX3_GPIO54
FX3_GPIO55
FPGA_CFG_SDIN
FX3_USBVBUS
FX3_I2C_SCL
FX3_I2C_SDA
GPIO[0]
GPIO[1]
GPIO[2]
GPIO[3]
GPIO[4]
GPIO[5]
GPIO[6]
GPIO[7]
GPIO[8]
GPIO[9]
GPIO[10]
GPIO[11]
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GPIO[39]
GPIO[40]
GPIO[41]/A0
GPIO[42]/A1
GPIO[43]
GPIO[44]
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GPIO[49]
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GPIO[53]
GPIO[54]
GPIO[55]
GPIO[56]
GPIO[57]
I2C_GPIO[58]
I2C_GPIO[59]
O[60]
VBATT
VBUS
DP
DM
SSTXP
SSTXM
SSRXP
SSRXM
R_USB2
R_USB3
INT#
TRST#
RESET#
TCK
TMS
TDI
TDO
CLKIN
CLKIN_32
XTALIN
XTALOUT
FSLC[0]
FSLC[1]
FSLC[2]
OTG_ID
FX3_FSD_P
FX3_FSD_N
FX3_SSTX_P
FX3_SSTX_N
FX3_SSRX_P
FX3_SSRX_N
FX3_INT
FX3_RESET
FX3_I2C_SCL
FX3_I2C_SDA
USB_CC1
USB_TX1_P
USB_TX1_N
USB_TX2_N
USB_TX2_P
USB_RX1_P
USB_RX1_N
USB_RX2_N
USB_RX2_P
REXT
Vconn_OC
Vconn_UVLO
High_Z
Current_Settingout1
CC1
TX1_P
TX1_N
TX2_N
TX2_P
RX1_P
RX1_N
RX2_N
RX2_P
CurrentSettingout
EP
CC2
Vconn_5Vin
VCC5V
CE
CurrentSettingin
CurrentSettingin
ORI_STATUS
TXP
TXN
GND
RX_P
RX_N
Attached_Status
CurrentSettingout
USB_CC2
FX3_SSTX_L_P
FX3_SSTX_L_N
FX3_SSRX_P
FX3_SSRX_N
Attached_Status
Current_Settingout0
A0
A1
A2
GND
VCC
WP
SCL
SDA
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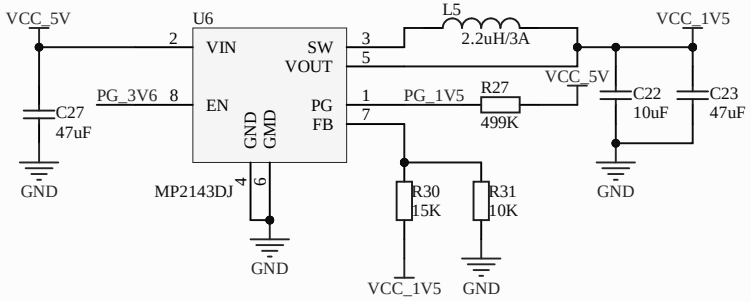

TYPE-C 24P QT
USB1
GND
VBUS
USB_TX2_P
USB_TX2_N
USB_CC2
FX3_FSD_P
FX3_FSD_N
USB_RX1_N
USB_RX1_P
USB_RX2_P
USB_RX2_N
USB_CC1
USB_TX1_N
USB_TX1_P
IC1B
CYUSB3014-BZXC
VCC1V8_FX3_CLK
VCC_1V8
VCC_1V2
VCC1V2_FX3_A
VCC1V2_FX3_RX
VCC1V2_FX3_TX
Attached_Status
R69
330R
C160
100nF
C161
10uF
C162
100nF
C163
1nF
C164
47uF
IC10
TPS22810DRVR
VCC_5V
C165
C166
C167
C168
C169
C170
C171
C172
C173
C174
C175
C176
C177
C178
C179
C180
C181
C182
L10
L11
L12
L13
L/120R
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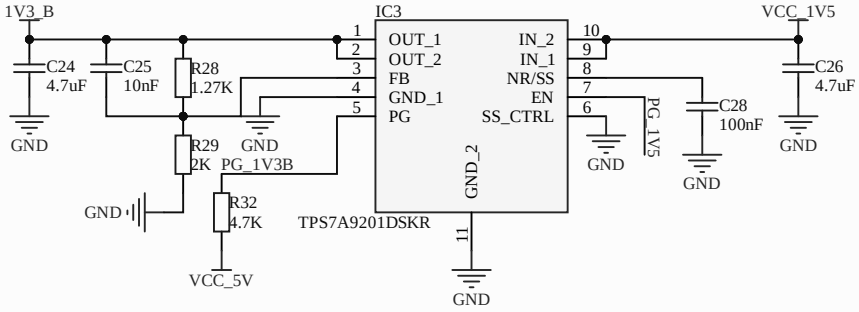


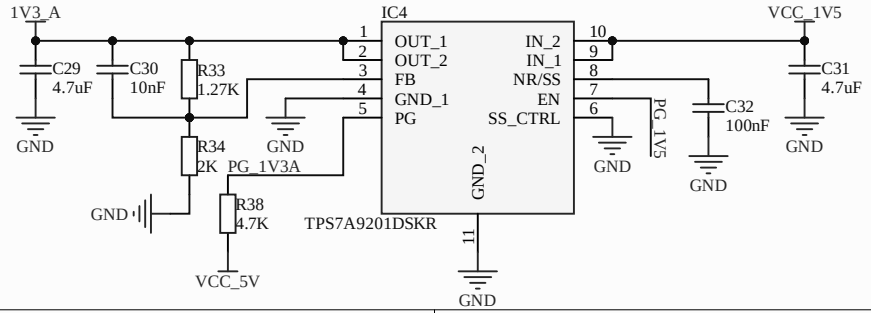


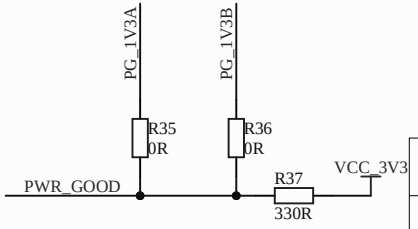




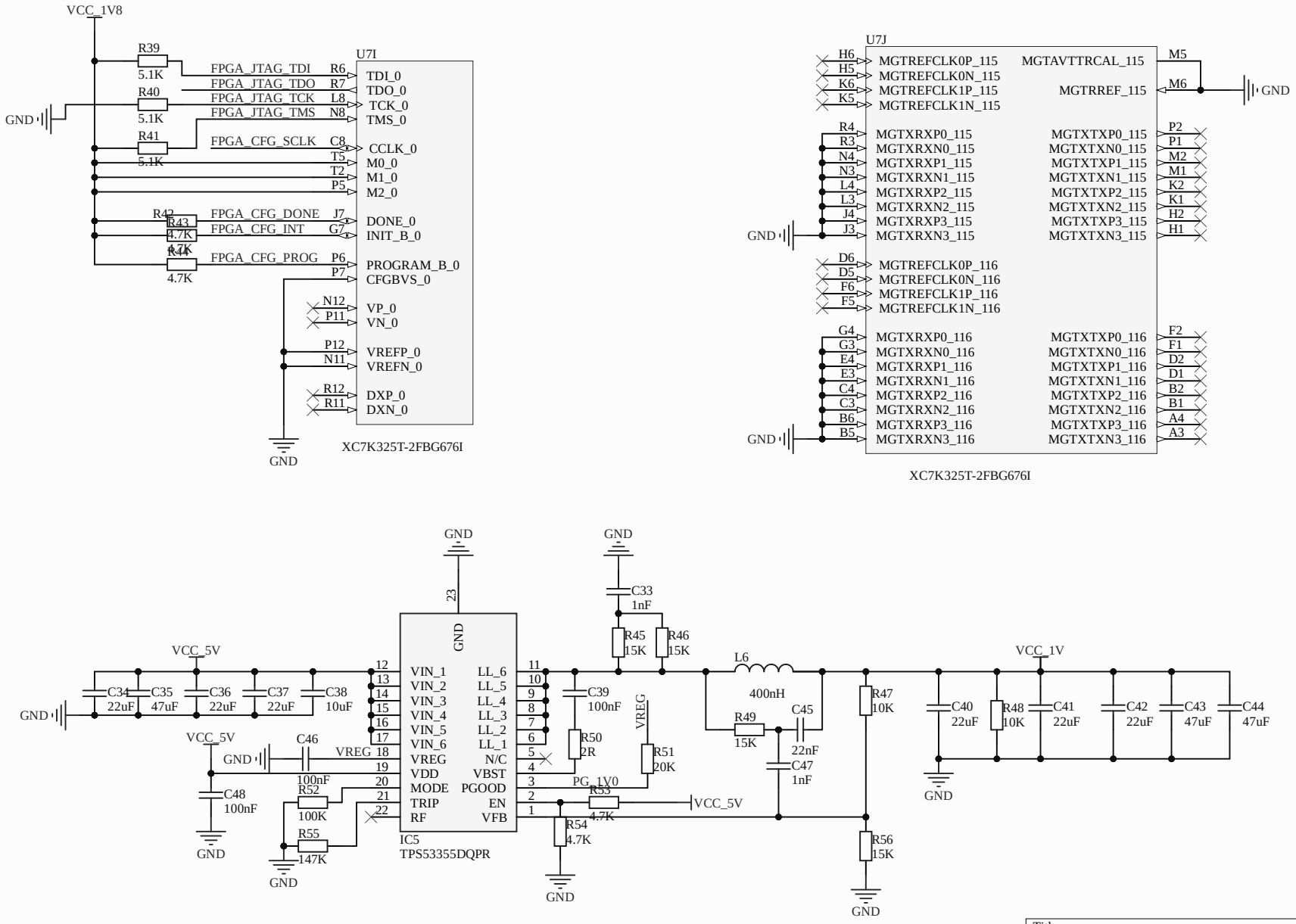




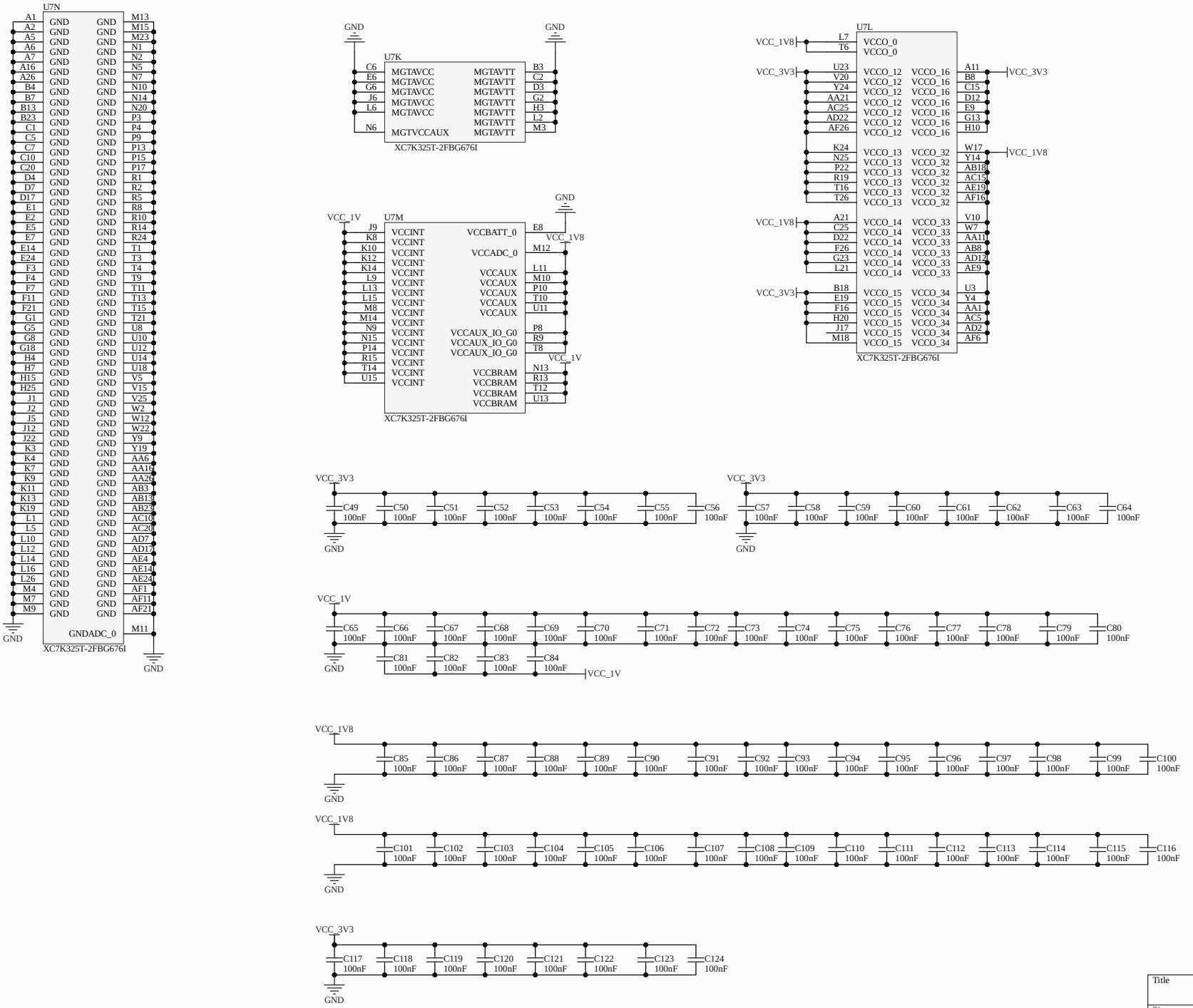


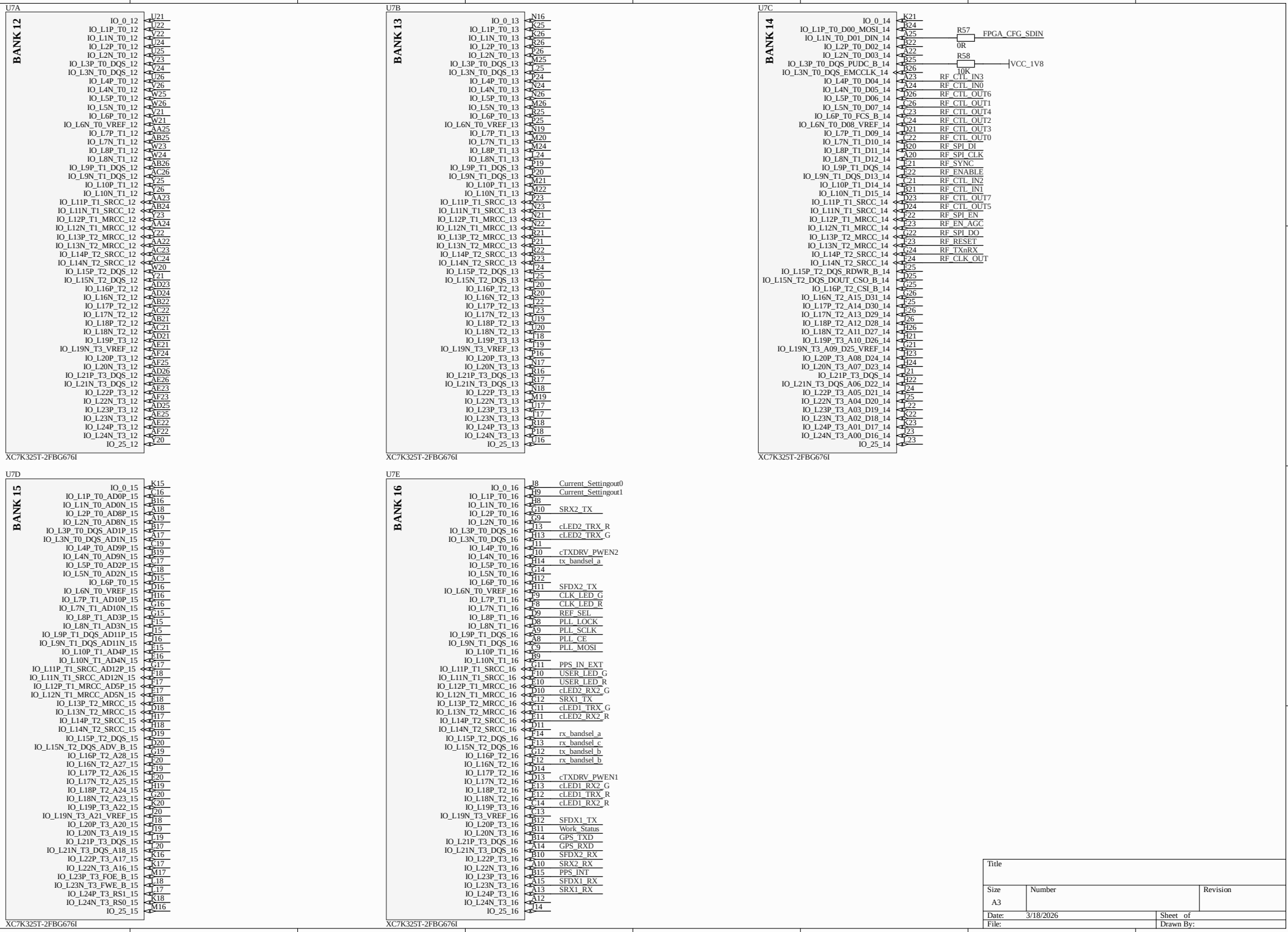


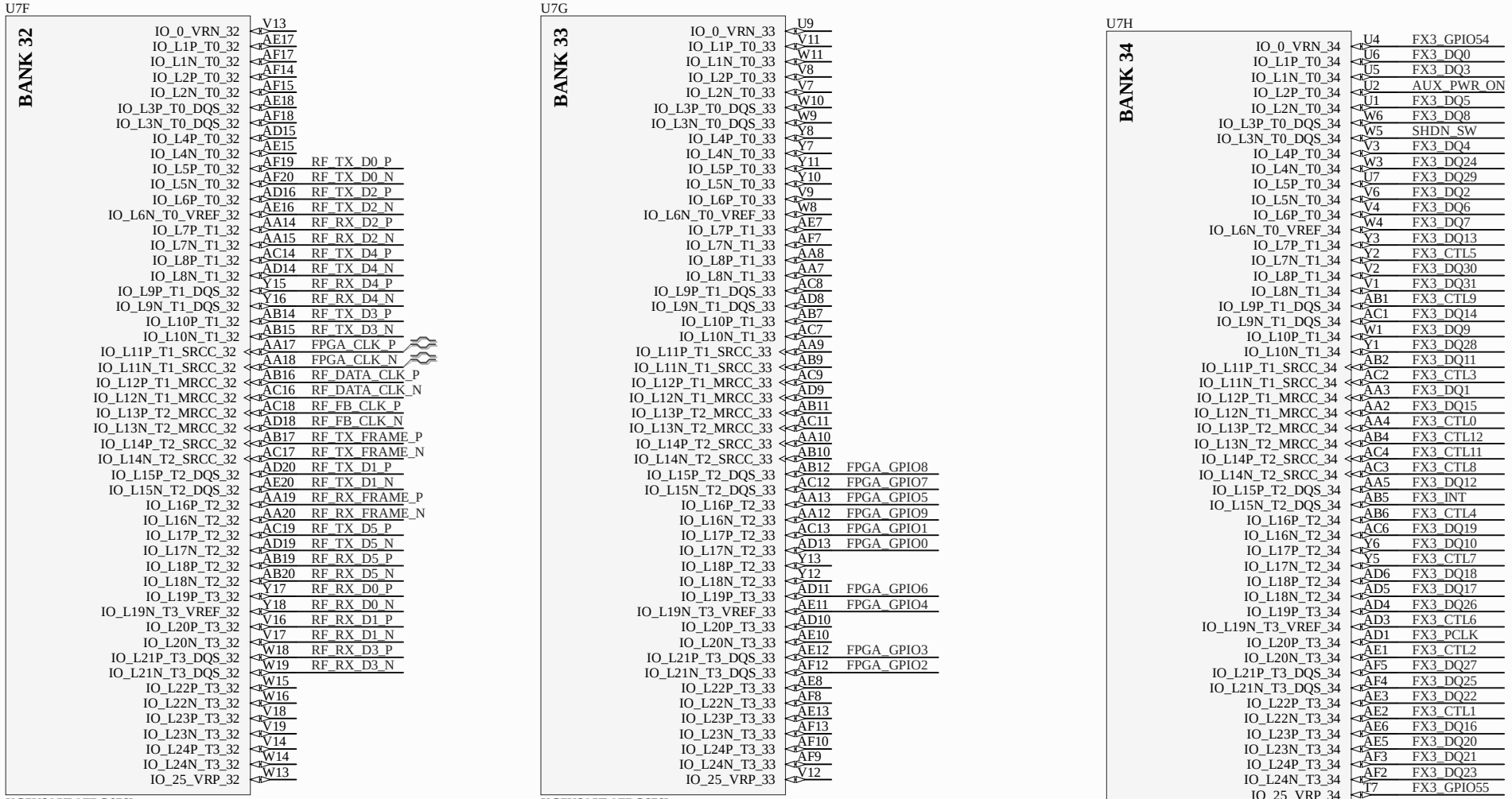
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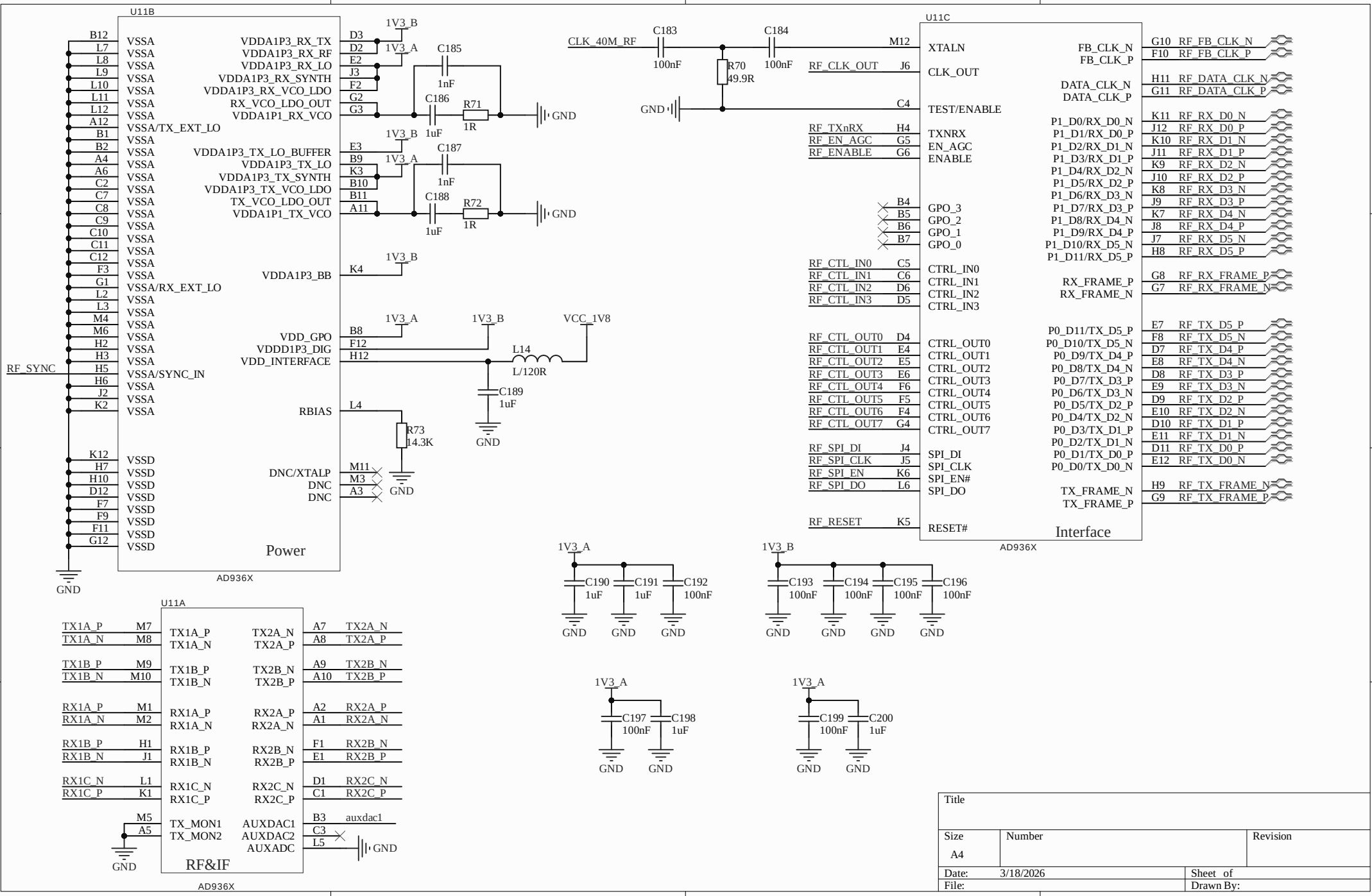


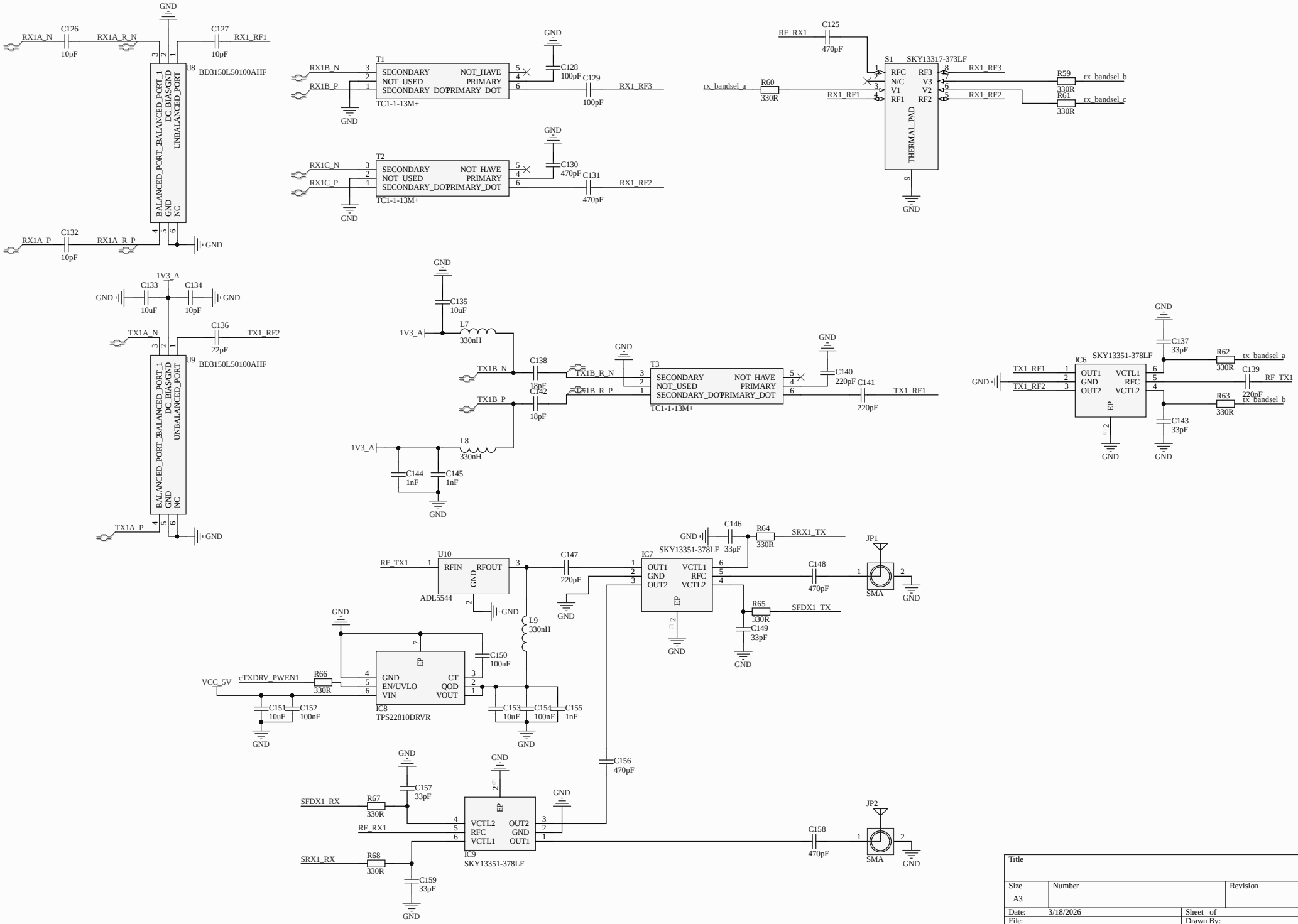
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U7K
XC7K325T-2FBG676I
U7M
XC7K325T-2FBG676I
U7L
XC7K325T-2FBG676I
GND
VCC_1V8
VCC_3V3
VCC_1V
C49 100nF
C50 100nF
C51 100nF
C52 100nF
C53 100nF
C54 100nF
C55 100nF
C56 100nF
C57 100nF
C58 100nF
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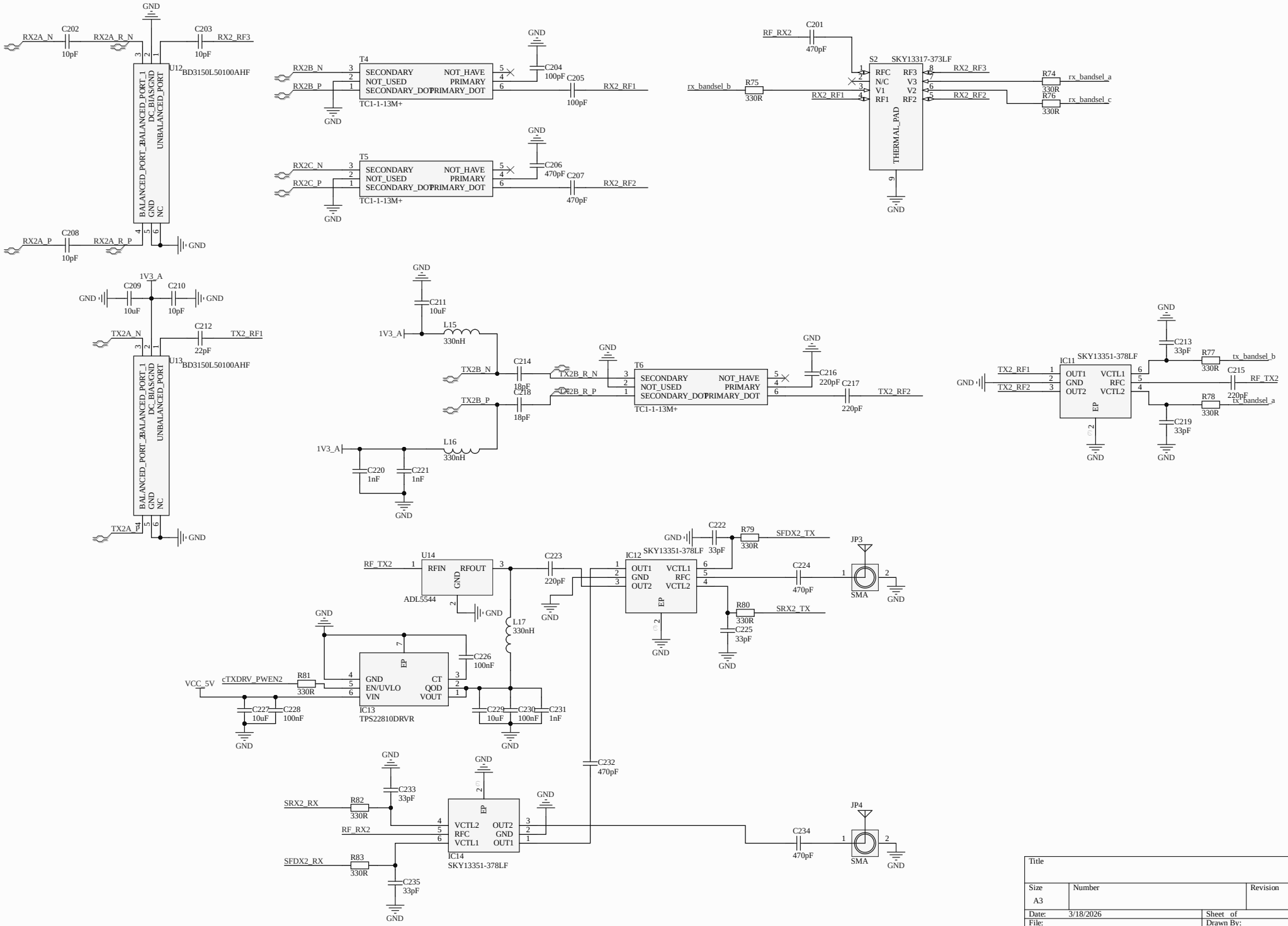

U7A
BANK 12
IO_0_12 U21
IO_L1P_T0_12 U22
IO_L1N_T0_12 V22
IO_L2P_T0_12 U24
IO_L2N_T0_12 U25
IO_L3P_T0_DQS_12 V23
IO_L3N_T0_DQS_12 V24
IO_L4P_T0_12 U26
IO_L4N_T0_12 V26
IO_L5P_T0_12 W25
IO_L5N_T0_12 W26
IO_L6P_T0_12 V21
IO_L6N_T0_VREF_12 W21
IO_L7P_T1_12 AA25
IO_L7N_T1_12 AB25
IO_L8P_T1_12 W23
IO_L8N_T1_12 W24
IO_L9P_T1_DQS_12 AB26
IO_L9N_T1_DQS_12 AC26
IO_L10P_T1_12 Y25
IO_L10N_T1_12 Y26
IO_L11P_T1_SRCC_12 AA23
IO_L11N_T1_SRCC_12 AB24
IO_L12P_T1_MRCC_12 Y23
IO_L12N_T1_MRCC_12 AA24
IO_L13P_T2_MRCC_12 Y22
IO_L13N_T2_MRCC_12 AA22
IO_L14P_T2_SRCC_12 AC23
IO_L14N_T2_SRCC_12 AC24
IO_L15P_T2_DQS_12 W20
IO_L15N_T2_DQS_12 Y21
IO_L16P_T2_12 AD23
IO_L16N_T2_12 AD24
IO_L17P_T2_12 AB22
IO_L17N_T2_12 AC22
IO_L18P_T2_12 AB21
IO_L18N_T2_12 AC21
IO_L19P_T3_12 AD21
IO_L19N_T3_VREF_12 AE21
IO_L20P_T3_12 AF24
IO_L20N_T3_12 AF25
IO_L21P_T3_DQS_12 AD26
IO_L21N_T3_DQS_12 AE26
IO_L22P_T3_12 AE23
IO_L22N_T3_12 AF23
IO_L23P_T3_12 AD25
IO_L23N_T3_12 AE25
IO_L24P_T3_12 AE22
IO_L24N_T3_12 AF22
IO_25_12 Y20
XC7K325T-2FBG676I
U7B
BANK 13
IO_0_13 N16
IO_L1P_T0_13 L25
IO_L1N_T0_13 L26
IO_L2P_T0_13 K26
IO_L2N_T0_13 J26
IO_L3P_T0_DQS_13 M25
IO_L3N_T0_DQS_13 L25
IO_L4P_T0_13 K24
IO_L4N_T0_13 K24
IO_L5P_T0_13 N26
IO_L5N_T0_13 M26
IO_L6P_T0_13 R25
IO_L6N_T0_VREF_13 P25
IO_L7P_T1_13 N19
IO_L7N_T1_13 M20
IO_L8P_T1_13 M24
IO_L8N_T1_13 L24
IO_L9P_T1_DQS_13 P19
IO_L9N_T1_DQS_13 P20
IO_L10P_T1_13 M21
IO_L10N_T1_13 M22
IO_L11P_T1_SRCC_13 P23
IO_L11N_T1_SRCC_13 N23
IO_L12P_T1_MRCC_13 R21
IO_L12N_T1_MRCC_13 R22
IO_L13P_T2_MRCC_13 R21
IO_L13N_T2_MRCC_13 P21
IO_L14P_T2_SRCC_13 R22
IO_L14N_T2_SRCC_13 R23
IO_L15P_T2_DQS_13 L24
IO_L15N_T2_DQS_13 K25
IO_L16P_T2_13 R20
IO_L16N_T2_13 R20
IO_L17P_T2_13 L22
IO_L17N_T2_13 L23
IO_L18P_T2_13 T19
IO_L18N_T2_13 T20
IO_L19P_T3_13 T18
IO_L19N_T3_VREF_13 T19
IO_L20P_T3_13 P16
IO_L20N_T3_13 N17
IO_L21P_T3_DQS_13 R16
IO_L21N_T3_DQS_13 R17
IO_L22P_T3_13 N18
IO_L22N_T3_13 M19
IO_L23P_T3_13 U17
IO_L23N_T3_13 L17
IO_L24P_T3_13 R18
IO_L24N_T3_13 P18
IO_25_13 U16
XC7K325T-2FBG676I
U7C
BANK 14
IO_0_14 K21
IO_L1P_T0_D00_MOSI_14 J24
IO_L1N_T0_D01_DIN_14 A25
IO_L2P_T0_D02_14 B22
IO_L2N_T0_D03_14 A22
IO_L3P_T0_DQS_PUDC_B_14 B25
IO_L3N_T0_DQS_EMCCLK_14 B26
IO_L4P_T0_D04_14 A23
IO_L4N_T0_D05_14 A24
IO_L5P_T0_D06_14 D26
IO_L5N_T0_D07_14 C26
IO_L6P_T0_FCS_B_14 C23
IO_L6N_T0_D08_VREF_14 C24
IO_L7P_T1_D09_14 D21
IO_L7N_T1_D10_14 C22
IO_L8P_T1_D11_14 B20
IO_L8N_T1_D12_14 A20
IO_L9P_T1_DQS_14 E21
IO_L9N_T1_DQS_D13_14 E22
IO_L10P_T1_D14_14 C21
IO_L10N_T1_D15_14 B21
IO_L11P_T1_SRCC_14 D23
IO_L11N_T1_SRCC_14 D24
IO_L12P_T1_MRCC_14 F22
IO_L12N_T1_MRCC_14 E23
IO_L13P_T2_MRCC_14 G22
IO_L13N_T2_MRCC_14 F23
IO_L14P_T2_SRCC_14 G24
IO_L14N_T2_SRCC_14 F24
IO_L15P_T2_DQS_RDWR_B_14 E25
IO_L15N_T2_DQS_DOUT_CSO_B_14 D25
IO_L16P_T2_CSI_B_14 G25
IO_L16N_T2_A15_D31_14 G26
IO_L17P_T2_A14_D30_14 F25
IO_L17N_T2_A13_D29_14 E26
IO_L18P_T2_A12_D28_14 J26
IO_L18N_T2_A11_D27_14 H26
IO_L19P_T3_A10_D26_14 H21
IO_L19N_T3_A09_D25_VREF_14 G21
IO_L20P_T3_A08_D24_14 J23
IO_L20N_T3_A07_D23_14 J24
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IO_L21N_T3_DQS_A06_D22_14 H22
IO_L22P_T3_A05_D21_14 J24
IO_L22N_T3_A04_D20_14 J25
IO_L23P_T3_A03_D19_14 L22
IO_L23N_T3_A02_D18_14 K22
IO_L24P_T3_A01_D17_14 K23
IO_L24N_T3_A00_D16_14 J23
IO_25_14 L23
R57 FPGA_CFG_SDIN
0R
R58 VCC_1V8
10K
RF_CTL_IN3
RF_CTL_IN0
RF_CTL_OUT6
RF_CTL_OUT1
RF_CTL_OUT4
RF_CTL_OUT2
RF_CTL_OUT3
RF_CTL_OUT0
RF_SPI_DI
RF_SPI_CLK
RF_SYNC
RF_ENABLE
RF_CTL_IN2
RF_CTL_IN1
RF_CTL_OUT7
RF_CTL_OUT5
RF_SPI_EN
RF_EN_AGC
RF_SPI_DO
RF_RESET
RF_TXnRX
RF_CLK_OUT
XC7K325T-2FBG676I
U7D
BANK 15
IO_0_15 K15
IO_L1P_T0_AD0P_15 C16
IO_L1N_T0_AD0N_15 B16
IO_L2P_T0_AD8P_15 A18
IO_L2N_T0_AD8N_15 A19
IO_L3P_T0_DQS_AD1P_15 B17
IO_L3N_T0_DQS_AD1N_15 A17
IO_L4P_T0_AD9P_15 C19
IO_L4N_T0_AD9N_15 B19
IO_L5P_T0_AD2P_15 C17
IO_L5N_T0_AD2N_15 C18
IO_L6P_T0_15 D15
IO_L6N_T0_VREF_15 D16
IO_L7P_T1_AD10P_15 H16
IO_L7N_T1_AD10N_15 G16
IO_L8P_T1_AD3P_15 G15
IO_L8N_T1_AD3N_15 F15
IO_L9P_T1_DQS_AD11P_15 J15
IO_L9N_T1_DQS_AD11N_15 J16
IO_L10P_T1_AD4P_15 E15
IO_L10N_T1_AD4N_15 E16
IO_L11P_T1_SRCC_AD12P_15 G17
IO_L11N_T1_SRCC_AD12N_15 F18
IO_L12P_T1_MRCC_AD5P_15 F17
IO_L12N_T1_MRCC_AD5N_15 E17
IO_L13P_T2_MRCC_15 E18
IO_L13N_T2_MRCC_15 D18
IO_L14P_T2_SRCC_15 H17
IO_L14N_T2_SRCC_15 H18
IO_L15P_T2_DQS_15 D19
IO_L15N_T2_DQS_ADV_B_15 D20
IO_L16P_T2_A28_15 G19
IO_L16N_T2_A27_15 F20
IO_L17P_T2_A26_15 F19
IO_L17N_T2_A25_15 E20
IO_L18P_T2_A24_15 H19
IO_L18N_T2_A23_15 G20
IO_L19P_T3_A22_15 K20
IO_L19N_T3_A21_VREF_15 J20
IO_L20P_T3_A20_15 J18
IO_L20N_T3_A19_15 J19
IO_L21P_T3_DQS_15 L19
IO_L21N_T3_DQS_A18_15 L20
IO_L22P_T3_A17_15 K16
IO_L22N_T3_A16_15 K17
IO_L23P_T3_FOE_B_15 M17
IO_L23N_T3_FWE_B_15 L18
IO_L24P_T3_RS1_15 L17
IO_L24N_T3_RS0_15 K18
IO_25_15 M16
XC7K325T-2FBG676I
U7E
BANK 16
IO_0_16 J8 Current_Settingout0
IO_L1P_T0_16 H9 Current_Settingout1
IO_L1N_T0_16 H8
IO_L2P_T0_16 G10 SRX2_TX
IO_L2N_T0_16 G9
IO_L3P_T0_DQS_16 J13 cLED2_TRX_R
IO_L3N_T0_DQS_16 H13 cLED2_TRX_G
IO_L4P_T0_16 J11
IO_L4N_T0_16 J10 cTXDRV_PWEN2
IO_L5P_T0_16 H14 tx_bandsel_a
IO_L5N_T0_16 G14
IO_L6P_T0_16 H12
IO_L6N_T0_VREF_16 H11 SFDX2_TX
IO_L7P_T1_16 F9 CLK_LED_G
IO_L7N_T1_16 F8 CLK_LED_R
IO_L8P_T1_16 D9 REF_SEL
IO_L8N_T1_16 D8 PLL_LOCK
IO_L9P_T1_DQS_16 A9 PLL_SCLK
IO_L9N_T1_DQS_16 A8 PLL_CE
IO_L10P_T1_16 C9 PLL_MOSI
IO_L10N_T1_16 B9
IO_L11P_T1_SRCC_16 G11 PPS_IN_EXT
IO_L11N_T1_SRCC_16 F10 USER_LED_G
IO_L12P_T1_MRCC_16 E10 USER_LED_R
IO_L12N_T1_MRCC_16 D10 cLED2_RX2_G
IO_L13P_T2_MRCC_16 C12 SRX1_TX
IO_L13N_T2_MRCC_16 C11 cLED1_TRX_G
IO_L14P_T2_SRCC_16 E11 cLED2_RX2_R
IO_L14N_T2_SRCC_16 D11
IO_L15P_T2_DQS_16 F14 rx_bandsel_a
IO_L15N_T2_DQS_16 F13 rx_bandsel_c
IO_L16P_T2_16 G12 tx_bandsel_b
IO_L16N_T2_16 F12 rx_bandsel_b
IO_L17P_T2_16 D14
IO_L17N_T2_16 D13 cTXDRV_PWEN1
IO_L18P_T2_16 E13 cLED1_RX2_G
IO_L18N_T2_16 E12 cLED1_TRX_R
IO_L19P_T3_16 C14 cLED1_RX2_R
IO_L19N_T3_VREF_16 C13
IO_L20P_T3_16 B12 SFDX1_TX
IO_L20N_T3_16 B11 Work_Status
IO_L21P_T3_DQS_16 B14 GPS_TXD
IO_L21N_T3_DQS_16 A14 GPS_RXD
IO_L22P_T3_16 B10 SFDX2_RX
IO_L22N_T3_16 A10 SRX2_RX
IO_L23P_T3_16 B15 PPS_INT
IO_L23N_T3_16 A15 SFDX1_RX
IO_L24P_T3_16 A13 SRX1_RX
IO_L24N_T3_16 A12
IO_25_16 J14
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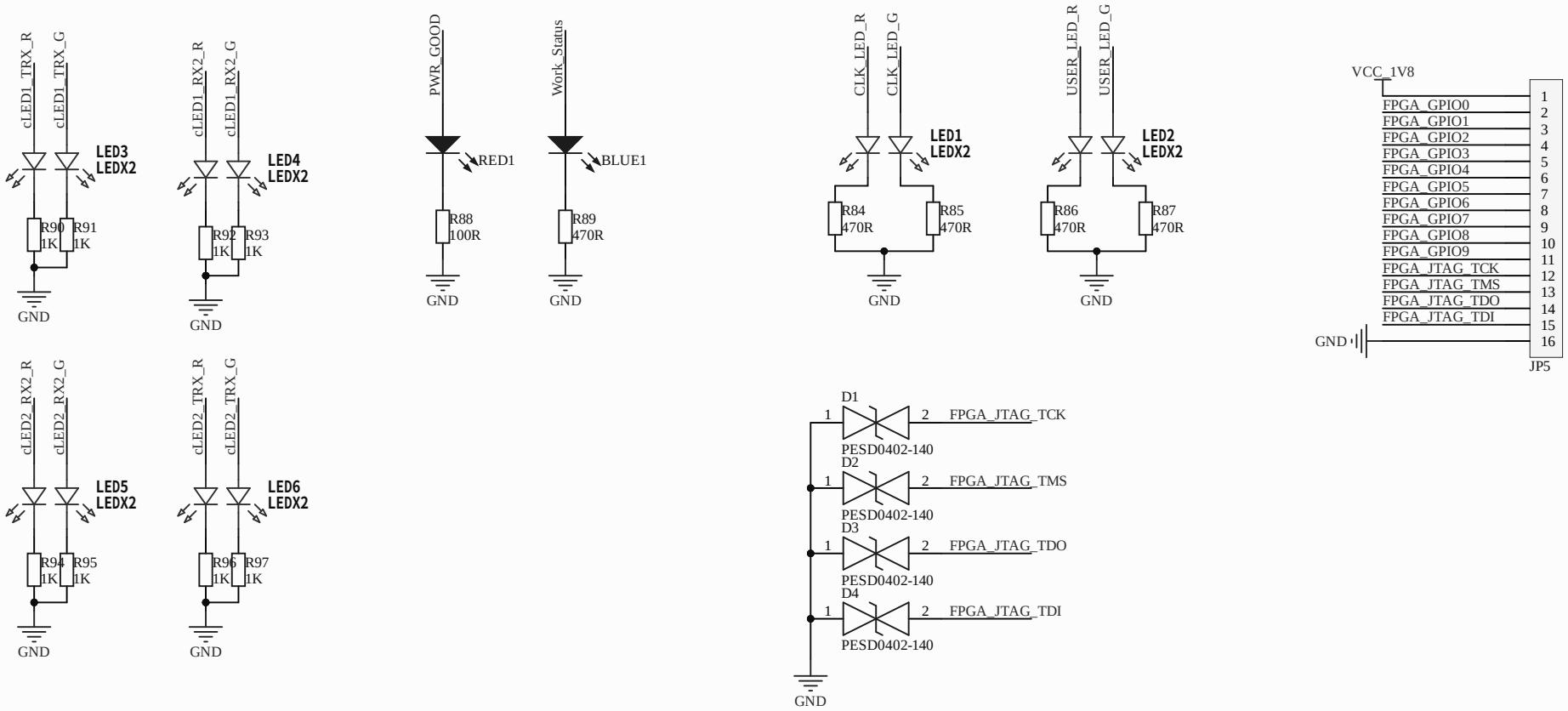

U7F
BANK 32
IO_0_VRN_32 V13
IO_L1P_T0_32 AE17
IO_L1N_T0_32 AF17
IO_L2P_T0_32 AF14
IO_L2N_T0_32 AF15
IO_L3P_T0_DQS_32 AE18
IO_L3N_T0_DQS_32 AF18
IO_L4P_T0_32 AD15
IO_L4N_T0_32 AE15
IO_L5P_T0_32 AF19 RF_TX_D0_P
IO_L5N_T0_32 AF20 RF_TX_D0_N
IO_L6P_T0_32 AD16 RF_TX_D2_P
IO_L6N_T0_VREF_32 AE16 RF_TX_D2_N
IO_L7P_T1_32 AA14 RF_RX_D2_P
IO_L7N_T1_32 AA15 RF_RX_D2_N
IO_L8P_T1_32 AC14 RF_TX_D4_P
IO_L8N_T1_32 AD14 RF_TX_D4_N
IO_L9P_T1_DQS_32 Y15 RF_RX_D4_P
IO_L9N_T1_DQS_32 Y16 RF_RX_D4_N
IO_L10P_T1_32 AB14 RF_TX_D3_P
IO_L10N_T1_32 AB15 RF_TX_D3_N
IO_L11P_T1_SRCC_32 AA17 FPGA_CLK_P
IO_L11N_T1_SRCC_32 AA18 FPGA_CLK_N
IO_L12P_T1_MRCC_32 AB16 RF_DATA_CLK_P
IO_L12N_T1_MRCC_32 AC16 RF_DATA_CLK_N
IO_L13P_T2_MRCC_32 AC18 RF_FB_CLK_P
IO_L13N_T2_MRCC_32 AD18 RF_FB_CLK_N
IO_L14P_T2_SRCC_32 AB17 RF_TX_FRAME_P
IO_L14N_T2_SRCC_32 AC17 RF_TX_FRAME_N
IO_L15P_T2_DQS_32 AD20 RF_TX_D1_P
IO_L15N_T2_DQS_32 AE20 RF_TX_D1_N
IO_L16P_T2_32 AA19 RF_RX_FRAME_P
IO_L16N_T2_32 AA20 RF_RX_FRAME_N
IO_L17P_T2_32 AC19 RF_TX_D5_P
IO_L17N_T2_32 AD19 RF_TX_D5_N
IO_L18P_T2_32 AB19 RF_RX_D5_P
IO_L18N_T2_32 AB20 RF_RX_D5_N
IO_L19P_T3_32 Y17 RF_RX_D0_P
IO_L19N_T3_VREF_32 Y18 RF_RX_D0_N
IO_L20P_T3_32 V16 RF_RX_D1_P
IO_L20N_T3_32 V17 RF_RX_D1_N
IO_L21P_T3_DQS_32 W18 RF_RX_D3_P
IO_L21N_T3_DQS_32 W19 RF_RX_D3_N
IO_L22P_T3_32 W15
IO_L22N_T3_32 W16
IO_L23P_T3_32 V18
IO_L23N_T3_32 V19
IO_L24P_T3_32 V14
IO_L24N_T3_32 W14
IO_25_VRP_32 W13
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U7G
BANK 33
IO_0_VRN_33 U9
IO_L1P_T0_33 Y11
IO_L1N_T0_33 W11
IO_L2P_T0_33 V8
IO_L2N_T0_33 V7
IO_L3P_T0_DQS_33 W10
IO_L3N_T0_DQS_33 W9
IO_L4P_T0_33 Y8
IO_L4N_T0_33 Y7
IO_L5P_T0_33 Y11
IO_L5N_T0_33 Y10
IO_L6P_T0_33 V9
IO_L6N_T0_VREF_33 W8
IO_L7P_T1_33 AE7
IO_L7N_T1_33 AF7
IO_L8P_T1_33 AA8
IO_L8N_T1_33 AA7
IO_L9P_T1_DQS_33 AC8
IO_L9N_T1_DQS_33 AD8
IO_L10P_T1_33 AB7
IO_L10N_T1_33 AC7
IO_L11P_T1_SRCC_33 AA9
IO_L11N_T1_SRCC_33 AB9
IO_L12P_T1_MRCC_33 AC9
IO_L12N_T1_MRCC_33 AD9
IO_L13P_T2_MRCC_33 AB11
IO_L13N_T2_MRCC_33 AC11
IO_L14P_T2_SRCC_33 AA10
IO_L14N_T2_SRCC_33 AB10
IO_L15P_T2_DQS_33 AB12 FPGA_GPIO8
IO_L15N_T2_DQS_33 AC12 FPGA_GPIO7
IO_L16P_T2_33 AA13 FPGA_GPIO5
IO_L16N_T2_33 AA12 FPGA_GPIO9
IO_L17P_T2_33 AC13 FPGA_GPIO1
IO_L17N_T2_33 AD13 FPGA_GPIO0
IO_L18P_T2_33 Y13
IO_L18N_T2_33 Y12
IO_L19P_T3_33 AD11 FPGA_GPIO6
IO_L19N_T3_VREF_33 AE11 FPGA_GPIO4
IO_L20P_T3_33 AD10
IO_L20N_T3_33 AE10
IO_L21P_T3_DQS_33 AE12 FPGA_GPIO3
IO_L21N_T3_DQS_33 AF12 FPGA_GPIO2
IO_L22P_T3_33 AE8
IO_L22N_T3_33 AF8
IO_L23P_T3_33 AE13
IO_L23N_T3_33 AF13
IO_L24P_T3_33 AF10
IO_L24N_T3_33 AF9
IO_25_VRP_33 Y12
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U7H
BANK 34
IO_0_VRN_34 U4 FX3_GPIO54
IO_L1P_T0_34 U6 FX3_DQ0
IO_L1N_T0_34 U5 FX3_DQ3
IO_L2P_T0_34 U2 AUX_PWR_ON
IO_L2N_T0_34 U1 FX3_DQ5
IO_L3P_T0_DQS_34 W6 FX3_DQ8
IO_L3N_T0_DQS_34 W5 SHDN_SW
IO_L4P_T0_34 V3 FX3_DQ4
IO_L4N_T0_34 W3 FX3_DQ24
IO_L5P_T0_34 U7 FX3_DQ29
IO_L5N_T0_34 V6 FX3_DQ2
IO_L6P_T0_34 V4 FX3_DQ6
IO_L6N_T0_VREF_34 W4 FX3_DQ7
IO_L7P_T1_34 Y3 FX3_DQ13
IO_L7N_T1_34 Y2 FX3_CTL5
IO_L8P_T1_34 V2 FX3_DQ30
IO_L8N_T1_34 Y1 FX3_DQ31
IO_L9P_T1_DQS_34 AB1 FX3_CTL9
IO_L9N_T1_DQS_34 AC1 FX3_DQ14
IO_L10P_T1_34 W1 FX3_DQ9
IO_L10N_T1_34 Y1 FX3_DQ28
IO_L11P_T1_SRCC_34 AB2 FX3_DQ11
IO_L11N_T1_SRCC_34 AC2 FX3_CTL3
IO_L12P_T1_MRCC_34 AA3 FX3_DQ1
IO_L12N_T1_MRCC_34 AA2 FX3_DQ15
IO_L13P_T2_MRCC_34 AA4 FX3_CTL0
IO_L13N_T2_MRCC_34 AB4 FX3_CTL12
IO_L14P_T2_SRCC_34 AC4 FX3_CTL11
IO_L14N_T2_SRCC_34 AC3 FX3_CTL8
IO_L15P_T2_DQS_34 AA5 FX3_DQ12
IO_L15N_T2_DQS_34 AB5 FX3_INT
IO_L16P_T2_34 AB6 FX3_CTL4
IO_L16N_T2_34 AC6 FX3_DQ19
IO_L17P_T2_34 Y6 FX3_DQ10
IO_L17N_T2_34 Y5 FX3_CTL7
IO_L18P_T2_34 AD6 FX3_DQ18
IO_L18N_T2_34 AD5 FX3_DQ17
IO_L19P_T3_34 AD4 FX3_DQ26
IO_L19N_T3_VREF_34 AD3 FX3_CTL6
IO_L20P_T3_34 AD1 FX3_PCLK
IO_L20N_T3_34 AE1 FX3_CTL2
IO_L21P_T3_DQS_34 AF5 FX3_DQ27
IO_L21N_T3_DQS_34 AF4 FX3_DQ25
IO_L22P_T3_34 AE3 FX3_DQ22
IO_L22N_T3_34 AE2 FX3_CTL1
IO_L23P_T3_34 AE6 FX3_DQ16
IO_L23N_T3_34 AE5 FX3_DQ20
IO_L24P_T3_34 AF3 FX3_DQ21
IO_L24N_T3_34 AF2 FX3_DQ23
IO_25_VRP_34 T7 FX3_GPIO55
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U11B
Power
AD936X
U11C
Interface
AD936X
U11A
RF&IF
AD936X
1V3_A
1V3_B
VCC_1V8
GND
RF_SYNC
CLK_40M_RF
C183 100nF
C184 100nF
R70 49.9R
C185 1nF
C186 1uF
R71 1R
C187 1nF
C188 1uF
R72 1R
L14 L/120R
C189 1uF
R73 14.3K
C190 1uF
C191 1uF
C192 100nF
C193 100nF
C194 100nF
C195 100nF
C196 100nF
C197 100nF
C198 1uF
C199 100nF
C200 1uF
VSSA
VSSD
VSSA/TX_EXT_LO
VSSA/RX_EXT_LO
VSSA/SYNC_IN
VDDA1P3_RX_TX
VDDA1P3_RX_RF
VDDA1P3_RX_LO
VDDA1P3_RX_SYNTH
VDDA1P3_RX_VCO_LDO
RX_VCO_LDO_OUT
VDDA1P1_RX_VCO
VDDA1P3_TX_LO_BUFFER
VDDA1P3_TX_LO
VDDA1P3_TX_SYNTH
VDDA1P3_TX_VCO_LDO
TX_VCO_LDO_OUT
VDDA1P1_TX_VCO
VDDA1P3_BB
VDD_GPO
VDDD1P3_DIG
VDD_INTERFACE
RBIAS
DNC/XTALP
DNC
XTALN
CLK_OUT
TEST/ENABLE
TXNRX
EN_AGC
ENABLE
GPO_3
GPO_2
GPO_1
GPO_0
CTRL_IN0
CTRL_IN1
CTRL_IN2
CTRL_IN3
CTRL_OUT0
CTRL_OUT1
CTRL_OUT2
CTRL_OUT3
CTRL_OUT4
CTRL_OUT5
CTRL_OUT6
CTRL_OUT7
SPI_DI
SPI_CLK
SPI_EN#
SPI_DO
RESET#
FB_CLK_N
FB_CLK_P
DATA_CLK_N
DATA_CLK_P
RX_FRAME_P
RX_FRAME_N
TX_FRAME_N
TX_FRAME_P
TX1A_P
TX1A_N
TX1B_P
TX1B_N
RX1A_P
RX1A_N
RX1B_P
RX1B_N
RX1C_N
RX1C_P
TX_MON1
TX_MON2
AUXDAC1
AUXDAC2
AUXADC
auxdac1
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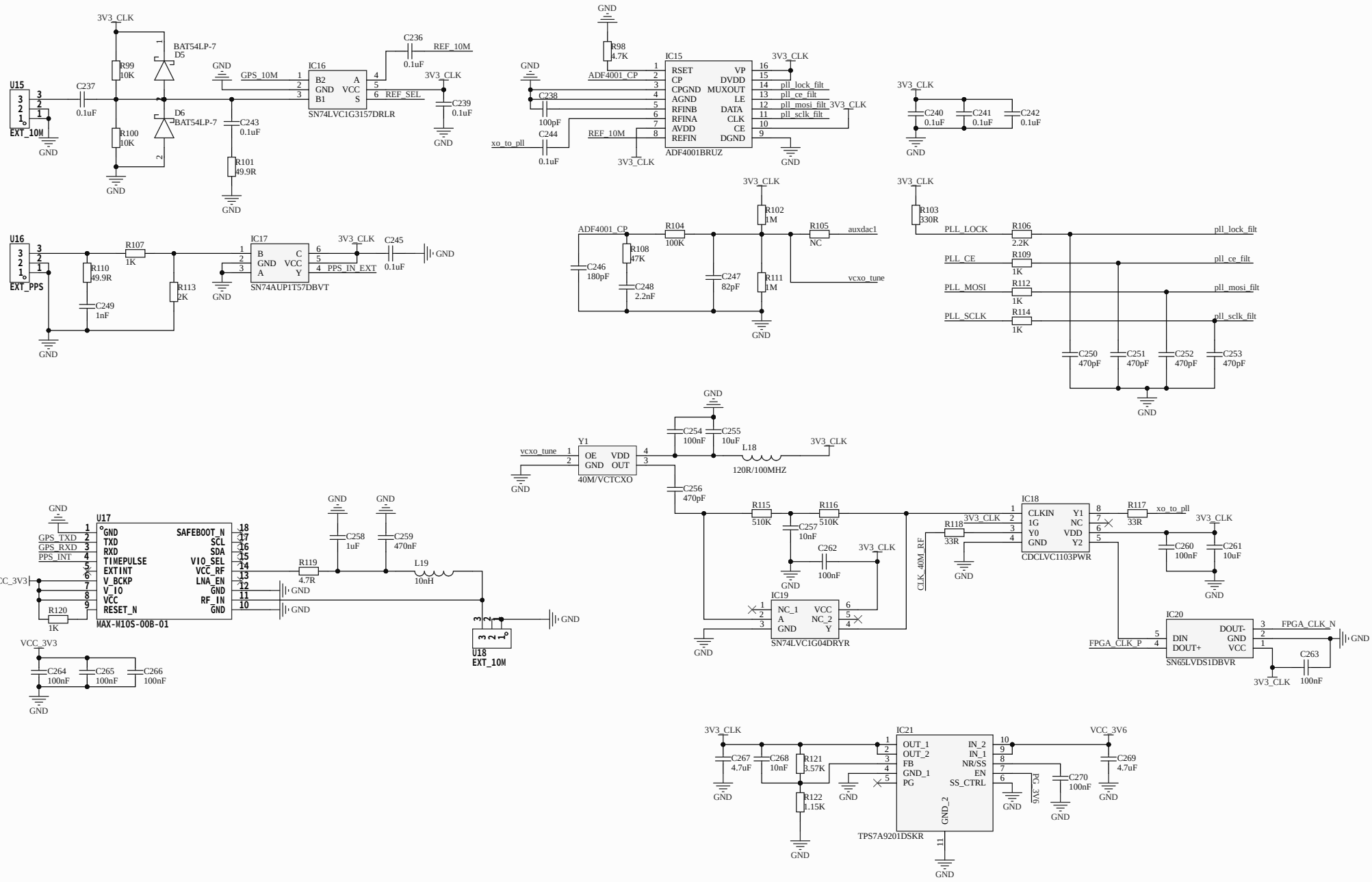


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GND
RX2A_N
C202
10pF
RX2A_R_N
C203
10pF
RX2_RF3
U12
BD3150L50100AHF
BALANCED_PORT_1
BALANCED_PORT_2
DC_BIAS/GND
UNBALANCED_PORT
GND
NC
RX2A_P
C208
10pF
RX2A_R_P
RX2B_N
RX2B_P
T4
SECONDARY
NOT_USED
SECONDARY_DOT
NOT_HAVE
PRIMARY
PRIMARY_DOT
TC1-1-13M+
C204
100pF
C205
100pF
RX2_RF1
RX2C_N
RX2C_P
T5
C206
470pF
C207
470pF
RX2_RF2
RF_RX2
C201
470pF
S2
SKY13317-373LF
RFC
N/C
V1
RF1
RF3
V3
V2
RF2
THERMAL_PAD
rx_bandsel_b
R75
330R
RX2_RF1
R74
330R
rx_bandsel_a
R76
330R
rx_bandsel_c
1V3_A
C209
10uF
C210
10pF
TX2A_N
C212
22pF
TX2_RF1
U13
BD3150L50100AHF
TX2A_P
C211
10uF
L15
330nH
TX2B_N
C214
18pF
TX2B_R_N
TX2B_P
C218
18pF
TX2B_R_P
T6
C216
220pF
C217
220pF
TX2_RF2
L16
330nH
C220
1nF
C221
1nF
IC11
SKY13351-378LF
OUT1
GND
OUT2
VCTL1
RFC
VCTL2
TX2_RF1
TX2_RF2
C213
33pF
R77
330R
tx_bandsel_b
C215
RF_TX2
R78
220pF
330R
tx_bandsel_a
C219
33pF
RF_TX2
U14
RFIN
RFOUT
ADL5544
C223
220pF
L17
330nH
IC12
SKY13351-378LF
C222
33pF
R79
330R
SFDX2_TX
C224
470pF
JP3
SMA
R80
330R
SRX2_TX
C225
33pF
C226
100nF
VCC_5V
cTXDRV_PWEN2
R81
330R
EP
GND
EN/UVLO
VIN
CT
QOD
VOUT
IC13
TPS22810DRVR
C227
10uF
C228
100nF
C229
10uF
C230
100nF
C231
1nF
C232
470pF
SRX2_RX
R82
330R
C233
33pF
RF_RX2
SFDX2_RX
R83
330R
C235
33pF
VCTL2
RFC
VCTL1
OUT2
GND
OUT1
IC14
SKY13351-378LF
C234
470pF
JP4
SMA
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